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INTRODUCTION TO SEWING MACHINE

1.1 INTRODUCTION

The sewing machine is a valuable tool in clothing construction. In older times, when there was no sewing machine, clothes were stitched by hand which was time consuming and the finished product was not so neat. The sewing machine was then invented in order to get the work done faster and neater.

Thomas Saint first used the wheel in 1790 in London. Things improved gradually and in 1846 Alice Hope invented a wooden machine. After many more inventions, Mr. Singer invented the metal sewing machine and a brand of machine is known by his name even today. The machine invented by Singer underwent many modifications till it came to the form we see it in today.

It is true that 'necessity is the mother of invention'. In order to facilitate work, many kinds of automatic sewing machines have been developed. Machines are of many kinds:-

1. Hand operated machines
2. Foot operated machines
3. Power operated machines (automatic machines)
4. Fashion maker, etc.

But before using a sewing machine, let us learn about the main parts of a machine. How to clean and maintain it? How to detect and rectify minor defects in it and also about the precautions to be observed during its use.

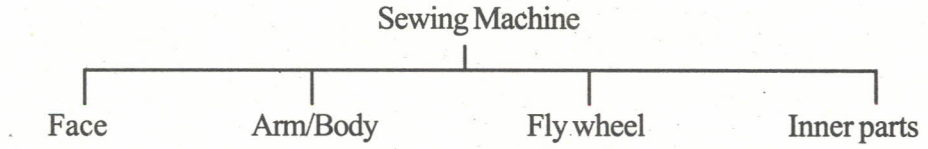
1.2 OBJECTIVES

After reading this lesson. You will be able to :-

- identify the main parts of a machine;
- state the precautions to be observed while using the machine;
- carry out appropriate maintenance of your machine;
- identify and correct the minor defects in your machine;
- use your machine properly.

1.3 PARTS OF THE MACHINE AND THEIR FUNCTIONS

Just like the human body is made up of many small and large bones, the sewing machine is also made up of many small parts. The machine can be divided into four main parts.



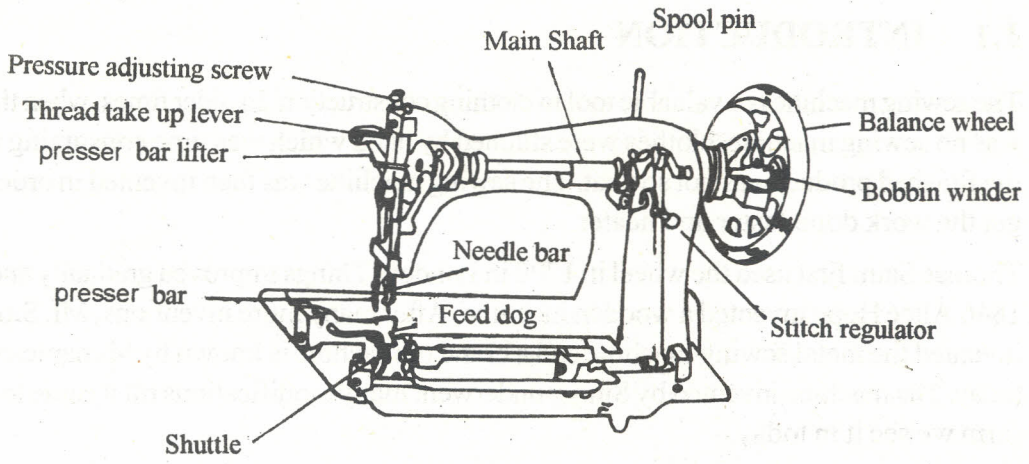


Fig. 1.1 Main parts of a machine

1. Face

This is the part where parts for threading, pressing and needle bar, etc., are located.

(i) Presser bar

This is the upright bar at the end of which the presser foot is attached.

(ii) Presser foot

This is a device for holding the material in place on the feed while stitching.

(iii) Presser foot lifter

This is the lever attached to the presser bar. It is used to lift or drop the presser foot on the material. It has to be lifted to remove the cloth from the machine.

(iv) Needle bar

This is the upright bar at the lower end of which the needle is attached. When the needle moves up and down, stitching takes place.

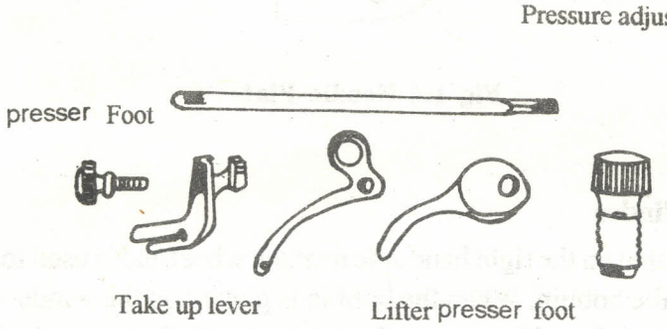


Fig. 1.2

(v) **Clamp** : It has a screw which is used to screw the needle into the proper place.

(vi) **Thread tension device** : This is a device on the bobbin case which regulates the tension of the bobbin thread.

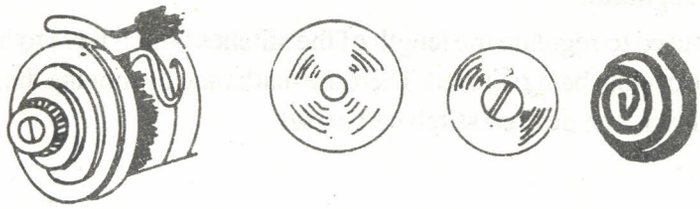


Fig. 1.3 Tension Discs

(vii) Take up lever

This is located above the tension regulator. At the outside end of the lever there is a small hole through which the thread passes. This lever performs two functions:-

- to feed the thread to the needle
- to tighten the loop formed by the shuttle.

As the lever moves up and down, it releases the thread to interlock with the bobbin thread, then as it rises, it tightens the thread to form a stitch.

(viii) Needle Plate

This is a semi-circular steel plate under the needle bar. It has a small hold through which the needle passes down to form a loop with the bobbin thread and come up.

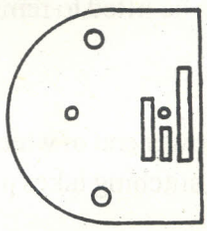


Fig 1.4 Needle Plate

2. Arm

(i) Bobbin Winder

This is located on the right hand side near the wheel and is used for winding the thread on the bobbin. When the bobbin is placed on the winder and the lever pressed down, the rubber ring makes contact with the nut on the fly wheel and the bobbin gets filled with thread.

(ii) Spool pin

These are two upright pins on the top of the arm which are used to hold the spools of thread so that it does not get tangled.

(iii) Stitch regulator

This is used to regulate the length of the stitches which have to be according to the type of cloth being stitched. There are marks and by moving the stitch regulator up or down, the desired stitch can be set.

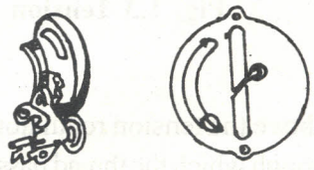


Fig. 1.5 (a) Bobbin winder (b) Stitch Regulator

3. Fly Wheel

(i) Balance wheel

This is the wheel at the center of the fly wheel and is the main part which drives the machine.

(ii) Fly wheel

This is the large wheel on the right hand side of the machine.

(iii) Stop motion screw or clamp

This is located near the center of the balance wheel. The machine does not work if the clamp is loosened. The screw or clamp is released or unscrewed while the bobbin is being wound.

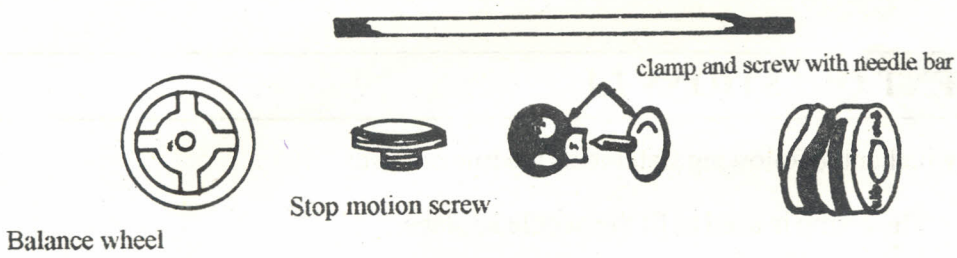


Fig. 1.6 (a) Balance wheel (b) stop motion screw

(iv) Bracket screw

This is present in hand operated machines. One end of this screw is attached to the balance wheel and the other end to the larger fly wheel.

4. Inner parts

(i) Bobbin

This is a small rod with two discs on both ends and the thread wound on this rod.

(ii) Bobbin case

The bobbin case holds the bobbin and fits in below the feed dog.

(iii) Shuttle

This is semi-circular in shape. There is a lever at the center on which the bobbin case fits into place.

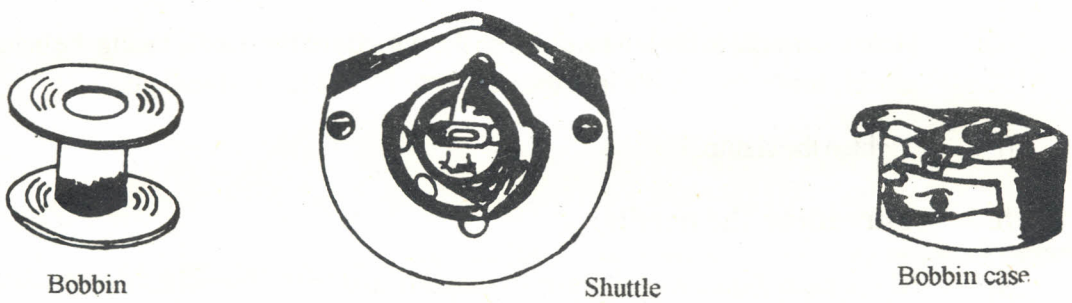


Fig.1.7 (a) Bobbin (b) Bobbin case (c) Shuttle

(iv) **Shuttle driver**

The lower part of the machine has a main shaft or rod which has a semi-circular part at one end. Over this part there is a spring which drives the shuttle.

(v) **Race body**

The lower part of the machine has a part within which the shuttle is kept and mounted on the shuttle driver with the help of two screws.

INTEXT QUESTIONS 1.1

State whether the following statements are true or false:

(a) The clamp is used to fit the needle in place.

(b) The bobbin winder is not used fill the bobbin.

(c) The bracket screw is used in foot operated machines.

(d) The thread reel is placed on the spool pin.

(e) The stitch regulator is used to control the length of the stitch.

1.4 KNOWING AND USING THE MACHINE

It is extremely important to know how to use your machine properly before attempting to do so. It is only when you know how to use it properly that you will be able to do so. If you follow the following steps, you will easily learn to use your machine correctly.

a. Setting the needle

(i) Turn the balance wheel until the needle bar is at the highest point.

(ii) Loosen the clamp.

(iii) Hold the needle in the left hand. Keep the flat side of the needle facing the balance wheel. Insert the needle into the needle holder as far as it will go.

(iv) Tighten the clamp.

b. Threading the needle

(i) Move the balance wheel towards yourself so that the thread take up lever is at the highest point.

(ii) Place the reel of thread on the spool pin.

(iii) Keep the right hand on the balance wheel and handle the thread with the left hadn.

(iv) Pass the thread through the eye of the needle from left to right.

(v) Leave about 3" to 4" of thread extending beyond the needle.

Note : If the needle is not threaded in the sequence given above, the thread will break repeatedly.

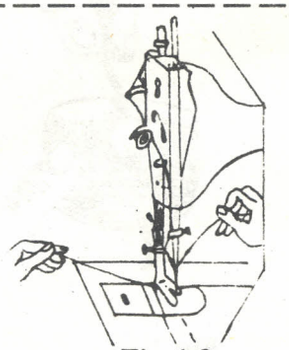


Fig 1.8

3. Winding the bobbin

(i) Release the stitching mechanism by loosening the stop motion screw.

(ii) Place the reel of thread on the spool pin.

(iii) Wind some thread on the bobbin and place it on the bobbin winder.

(iv) Press the lever down so that the rubber ring comes in contact with the fly wheel.

(v) Run the machine at a constant speed to ensure even winding of thread on the bobbin.

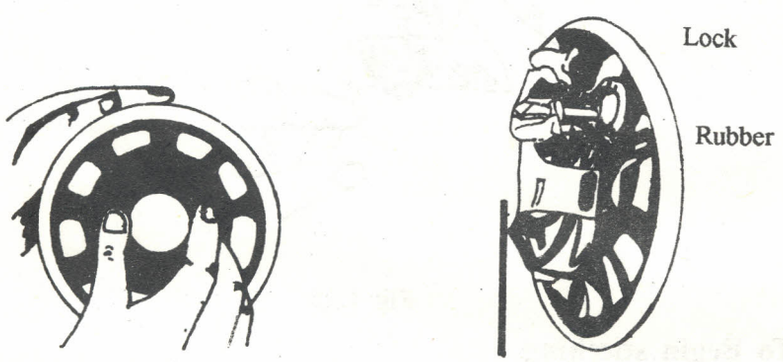


Fig 1.9

4. Placing the bobbin case

(i) Place the bobbin in the bobbin case so that the thread remains towards you.

(ii) Bring about 3" of thread out from the open part of the bobbin case.

(iii) Hold the ready bobbin case by opening the holding shaft of the shuttle and release the holding pin.

(iv) You will hear a click when the bobbin case fits in properly.

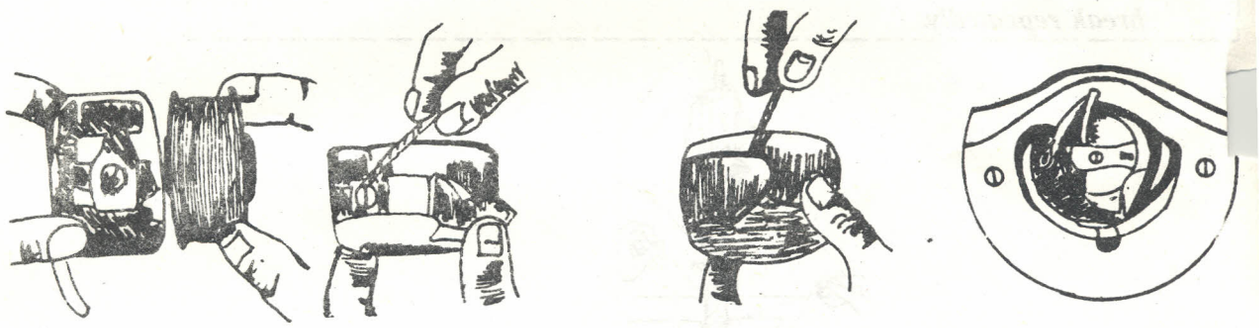


Fig 1.10

5. Bringing bobbin thread up

(i) Hold the needle thread in your left hand.

(ii) Move the fly wheel slowly towards yourself.

(iii) Let the needle go down, releasing the thread gradually. Then bring the needle up.

(iv) The upper thread will come up bringing the bobbin thread with it.

(v) Move both threads backwards under the presser foot.

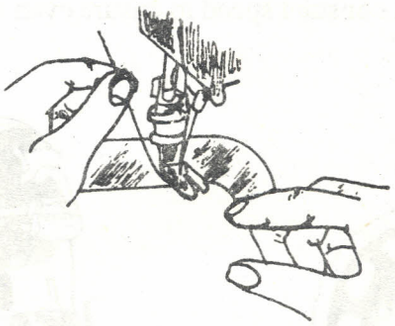


Fig 1.11

6. To Begin stitching

(i) First, bring the bobbin thread up.

(ii) Place the cloth to be stitched under the presser foot. Guide the edge of the cloth with your left hand.

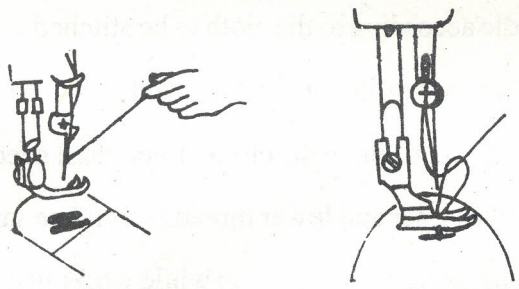


Fig. 1.12

(iii) Move the fly wheel towards yourself and start stitching. Feed the cloth with your left hand.

(iv) Cut the thread, leaving 2" extra, after completing the stitching.

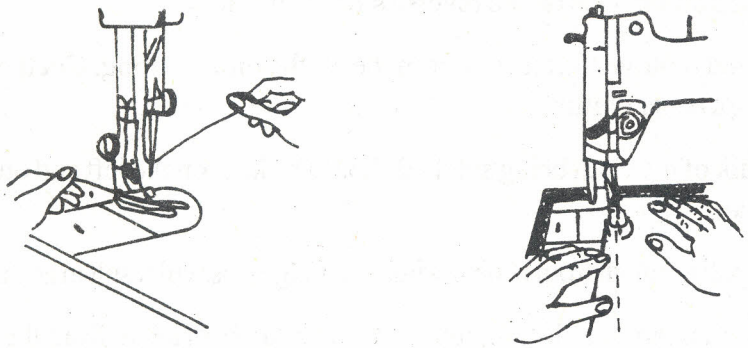


Fig. 1.13

INTEXT QUESTIONS 1.2

State whether the following statements are true or false.

(a) The thread winds on the bobbin when the fly wheel is tightened.

(b) The needle should be threaded in the proper sequence.

(c) The flat side of the needle should be on the outside.

(d) One should leave about 2" of thread after completing the stitching.

(e) There is no noise on fitting the bobbin case on the shuttle.

1.5 PRECAUTIONS WHILE USING A MACHINE.

1. Keep the machine clean both from inside and outside.
2. Select the needle according to the cloth to be stitched.
3. Check to see that the needle is properly fitted.
4. Select the thread according to the cloth - finer cloth needs a finer thread.
5. The tension of the upper and lower threads should be same.
6. One should sit up straight and not soop while working on the machine.
7. Light should fall from the left side of the machine.
8. There should be about 10 to 12 cm of thread - both upper and lower before beginning to stitch.
9. Cut all the parts of the garment before beginning to stitch. This will save your getting up again and again.
10. Mark the stitch lines on the reverse side of the cloth.
11. Press and remove the creases from the cloth before cutting. Cut the cloth after making a paper drafting.
12. The bulk of the cloth being stitched should be kept on the left and outside of the machine.
13. Never pull the cloth from behind while stitching. This will result in an uneven seam.
14. While stitching, the cloth should be held about 8 cm away from the needle.
15. Use a basting stitch to mark the stitch line on woollen and printed material.
16. When starting or ending a seam, secure the ends.
17. Lift the presser foot up, remove the cloth by pulling it out slightly and then cut the thread, leaving about 8 cm spare.
18. The same thread should be used on the upper as well as lower side.
19. The thread used should be tangle free.
20. A hand operated sewing machine should be placed at least 30 cm above the ground.
21. Place a thin sheet of paper under delicate fabrics like net, georgette, etc. This makes the stitching neat and the paper can be removed after stitching.

22. When stitching denim or similar thick material, soak the cloth in soapy water for sometime or rub wax or soap over areas to be stitched. This makes stitching easier.

23. When fitting the bobbin case on the shuttle, listen for the click which indicates that it has been set properly.

24. When stitching a plain edge to a bias edge, keep the bias edge on top and baste it before stitching. This prevents puckering.

25. Keep the machine at an appropriate height and use a stool which allows you to rest your feet comfortably on the ground.

Note : While stitching by hand, keep the thread 27" to 30" long.

1.6 CLEANING AND MAINTENANCE OF A SEWING MACHINE

The sewing machine can be compared to a human body. Just as the human body needs adequate nutrition in order to remain healthy, the sewing machine, too needs to be oiled and cleaned regularly. Hence, we should learn how to maintain and oil the sewing machine. For this, we should keep the following points in mind.

(a) Cleaning the machine

A small brush and a soft cloth are needed for cleaning a sewing machine. After brushing with the brush, the dust is wiped off with a soft cloth. The machine has to be cleaned in two ways.

(i) cleaning the outside

(ii) cleaning the inside

(i) **Cleaning the outside** : The machine should be wiped with a soft cloth before beginning any work. This keeps the machine looking beautiful and the cloth being stitched does not get dirty.

(ii) **Cleaning the inside** : The inside of the machine should be carefully cleaned. After continuous use of the machine, a lot of lint and pieces of thread accumulate in the shuttle. This makes the machine heavy and it does not run smoothly. Sometimes, the machine may even get jammed. In order to keep it running smoothly, it must be properly cleaned from time to time.

Shuttle

Open the shuttle, clean it with a brush or a needle (if needed), wipe with a cloth and then

screw back into place.

Face plate

Open the face plate, clean the parts and oil them.

Teeth/feed dog

Unscrew the needle plate and clean the feed dog thoroughly by removing all lint, pieces of thread, dust and dirt.

Note : In order to clean the machine, all the parts should not be opened at the same time. Without adequate knowledge, it becomes difficult to put the parts back in their proper place.

(b) Oiling the machine

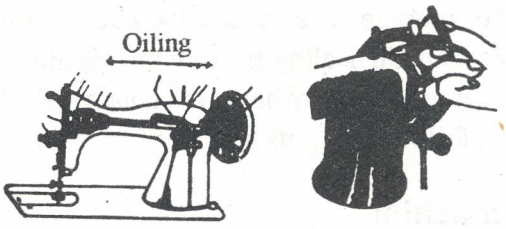


Fig 1.14

(i) The machine should always be oiled after cleaning. Remember, oil is food for the machine.

(ii) Whenever two parts rub against one another, there is friction and it is essential to oil such parts in order to minimize wear and tear.

(iii) There are 11 holes above the machine and 16 holes on the inside. These holes are meant for oiling. Leave the machine idle for sometime after oiling. Run the machine on a rough cloth first so that your garment does not get stained by oil.

(iv) People using the sewing machine from 10 to 12 hours daily, need to oil their machine every day. If the machine is used for about 3-4 hrs a week, the machine needs to be oiled once a week. When a machine is used only occasionally, it must be oiled at least once a month.

(v) Always use an oil of good quality.

(vi) Always make sure to use specially formulated machine oil only. Using any other oil like kerosene or mustard oil damages the parts and causes faster wear and tear.

(c) Proper maintenance of the machine

In addition to cleaning and oiling the machine, it is also essential to know how to maintain a machine.

(i) Keep it covered

After finishing your work, cover the machine carefully. If the cover has a lock, lock it. This will not only keep the dust out but will also prevent children from tampering with the machine.

(ii) Keep it away from children

Children should be kept away from the machine. They may unknowingly put their hand under the needle which may hurt them badly.

(iii) Keep it dry

Always try to keep the machine in a dry place, away from moisture. Moisture may cause rusting in the parts which will cause faster wear tear. The outer wooden box may also get distorted by moisture and dampness. Hence, it is essential to keep the machine in a dry and safe place.

(iv) Lossen the stop motion screw

The stop motion screw should always be loosened after finishing work and before putting away the machine. The machine does not move after lossening the stop motion screw so there is no danger of the thread entangling or the needle breaking even if a child happens to move the fly wheel.

(v) Lower the presser foot

While putting the machine away, always lower the presser foot as this takes the tension away from the spring.

(vi) Remove the thread from the needle

The thread should always be removed from the needle before storing so that there is no danger of the thread breaking and getting tangled if the machine is accidentally run.

ACTIVITY

Clean and oil your sewing machine according to the directions given above.

1.7 SOME COMMON DEFECTS

While using a sewing machine, one may come across many small defects. With appropriate knowledge, these defects can easily be identified and corrected at home without having to call a mechanic. You may come across the following defects:

(a) Machine working heavily

	Reason	Corrections
i)	Thread entangled in the shuttle.	Open the shuttle and remove the thread.
ii)	Bobbin winder may be pressed down.	Release the bobbin winder.
iii)	Some parts may be rusted	Replace rusted parts.
iv)	Machine not oiled for a long time.	Clean and oil the machine.
v)	Tension too light.	Adjust tension

(b) Needle breaking

	Reason	Corrections
i)	Needle not fitted properly.	Fit the needle properly.
ii)	Needle not appropriate for material.	Use proper size needle for the material being stitched.
iii)	Needle of inferior quality.	Use needle of good quality only.
iv)	The screw of the presser foot may be lose.	Tighten the screw.
v)	Bobbin case not set properly.	Remove and set the bobbin case again.
vi)	Needle may be bent while pulling the thread to cut it.	Use scissors to cut the thread.
vii)	Stitching unshrunk material.	Shrink the material before stitching.
viii)	Needle may be bent or crooked.	Replace the needle.
ix)	Shaft of the needle holder may be worn away.	Replace the needle holder shaft.

(c) Irregular stitch

	Reason	Corrections
i)	Upper and lower threads of different thickness.	Use the same thread.
ii)	Bobbin case worn out.	Replace the bobbin case.
iii)	Tension of upper or lower thread too loose or too light.	Adjust tension, keeping tension of lower thread slightly more loose than the upper one.
iv)	Needle set insecurely	Set the needle properly
v)	Needle blunt	Replace the needle.

(d) Thread breaking

	Reason	Corrections
i)	Thread used unsuitable to material being stitched.	Select thread according to material being stitched.
ii)	Shuttle wrongly placed or otherwise damaged.	Place shuttle correctly or replace if needed.
iii)	Upper thread incorrectly threaded	Thread properly.
iv)	Bobbin wound too tight or too full.	Rewind the bobbin properly.
v)	Needle incorrectly set	Remove and set correctly.
vi)	Tangled thread	Use smooth, tangel-free thread.
vii)	Thread entangled in spool pin.	Remove and clear the tangle.

(e) Material puckering

	Reason	Corrections
i)	Too much or too little pressure on the presser foot	Adjust to optimum pressure.
ii)	Tension of thread too light	Adjust tension.

iii)	Blunt needle	Replace needle
iv)	Thread not as fine as material being stitched.	Use suitable thread
v)	Presser foot damaged	Replace presser foot.

(f) Looping

	Reason	Corrections
i)	Loosely wound bobbin	Rewind bobbin correctly.
ii)	Incorrect upper and lower thread tension.	Adjust tension on both upper and lower threads.
iii)	Dust in the bobbin case and feed dog.	Clean and remove dust.
iv)	Incorrect upper and lower threading.	Redo the threading

(g) Stitch missing

	Reason	Corrections
i)	Incorrectly set needle	Remove and set again
ii)	Needle set too high or too low	Set needle correctly.
iii)	Material heavily starched	Destarch material before stitching.
iv)	Incorrect thread being used	Use suitable thread.
v)	Needle bent or blunt	Replace needle.

1.8 WHAT HAVE YOU LEARNT

In this lesson you have learnt about the main parts of a sewing machine and their functions, how to use a machine; the precautions to be observed while using a machine; how to clean, oil and maintain a machine and also to detect and correct the common defects of a sewing machine.

1.9 TERMINAL QUESTIONS

Q.1 What are the precautions one must observe while using a sewing machine?

Q.2 Describe in detail the process of cleaning, oiling and maintaining a sewing machine.

ACTIVITY

Observe the following defects in a sewing machine and suggest ways to correct them.

(a) Machine running heavy. (b) Needle breaking

(c) Looping

1.10 Answers to Intext Questions

1.1 (a) True (b) False (c) False

(d) True (e) True

1.2 (a) False (b) True (c) False

(d) True (e) False.